

Verify Data Quality

What is data quality verification?

Data verification includes checking data for:

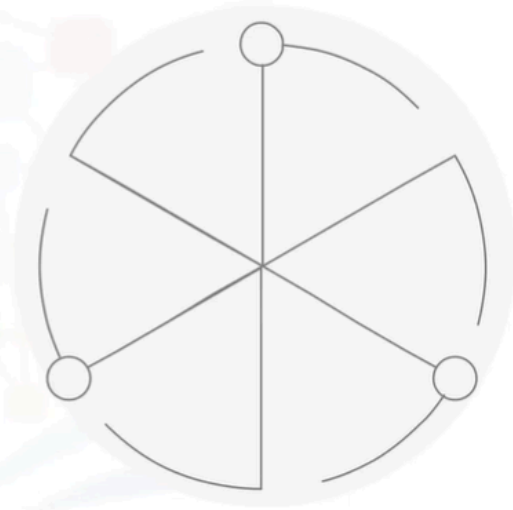


- Data verification includes checking your data for: Accuracy— Is your data correct? Completeness—Is there missing data? Consistency— Are fields consistently entered? and, Currency—Is your data up to date?

Why use data quality verification?

Data verification is about:

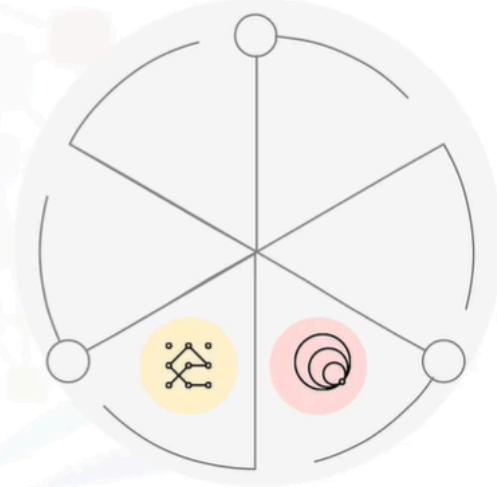
- Managing data quality
- Enhancing data reliability



Why use data quality verification?

High-quality data enables:

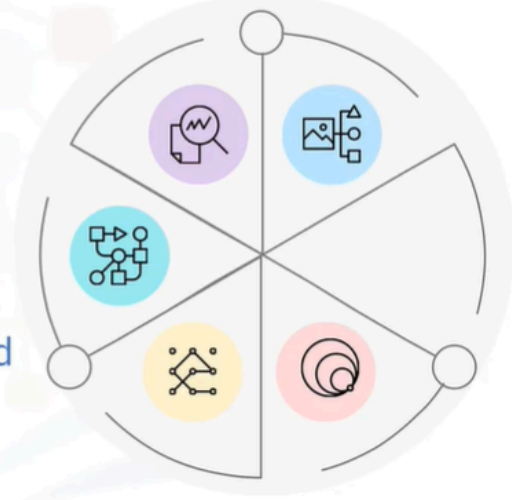
- Successful integration of data
- Integrity of complex relationships



Why use data quality verification?

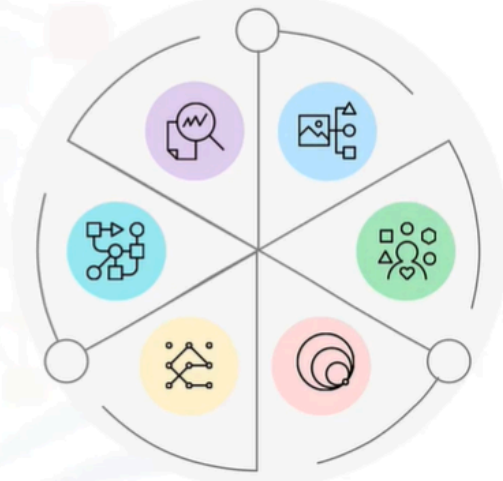
Data verification provides you with:

- A complete, connected view of your organization
- Data ready for advanced analysis
- Statistical modeling and machine learning



Why use data quality verification?

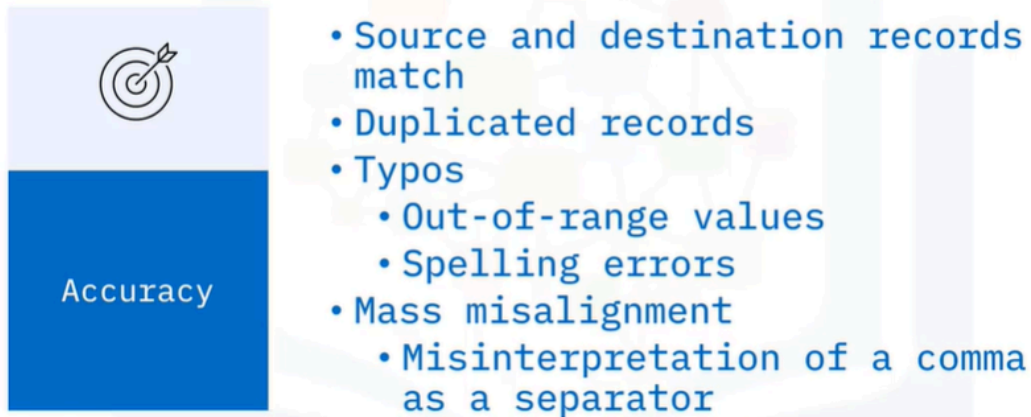
Boosts
confidence in
your insights
and decision-
making



- Data verification is about managing data quality and enhancing its reliability. High-quality data enables successful integration of related data and its complex relationships.
- Data verification also provides you with a complete and connected view of your organization, data that is ready for advanced analysis, statistical modeling and machine learning, and ultimately, more confidence in your insights and decision-making.

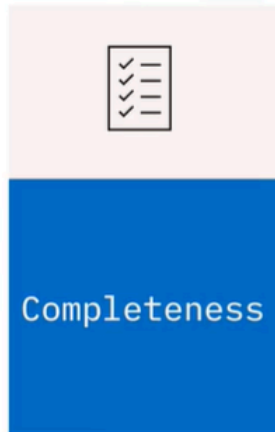
- Unfortunately, data quality is not a top concern among the daily chaos of running a company.

Data quality concerns



- The first is accuracy. Accuracy includes ensuring a match between source data and destination data. How can accuracy become an issue? Data migrating from source systems often contains duplicated records. When users enter data manually, typos can find their way into the data records, yielding out-of-range values, outliers, and spelling mistakes. Sometimes large chunks of data become misaligned, causing data corruption.
- For example, a CSV file might contain a legitimate comma, which the new system can misinterpret as a column separator.

Data quality concerns



- Locating missing values
 - Void or null fields
 - Placeholders like "999" or "-1"
- Locating missing data records due to system failures

- Another data quality concern is completeness. Data is incomplete when the business finds missing values, such as voids or nulls in fields that should be populated, or haphazard use of placeholders such as "9 9 9" or "minus 1" to indicate a missing value. Entire records can also be missing due to upstream system failures.

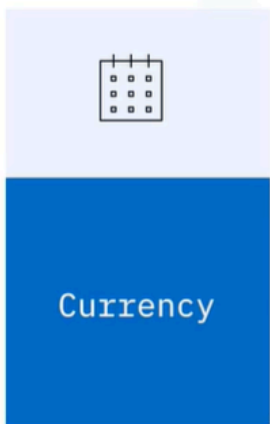
Data quality concerns



- Non-conformance to standard terms
- Date formatting
 - YMD and MDY
- Inconsistent data entry
 - "Mr. John Doe" and "John Doe"
- Inconsistent units
 - Metric and imperial
 - Dollars and thousands of dollars

- Consistency is another important data quality concern. Are there deviations from standard terminology? Are dates entered consistently? For example, year-month-day and month-day-year formats are incompatible.
- Is data entered consistently? For example, Mr. John Doe and John Doe might refer to the same person in the real world, but the system will see them as distinct. Are the units consistent? For example, you are expecting "kilograms," but you might have entries based on "pounds," or you are expecting 'dollar amounts,' but you might have entries based on "thousands of dollars."

Data quality concerns



- Avoiding outdated information
 - Update addresses: Check against a 'change of address' database
 - Manage name changes

- Lastly, currency is an ongoing data quality concern for most businesses. Currency is about ensuring your data remains up to date. For example, you might have dimension tables that contain customer addresses, some of which might be outdated. In the US, you could check these against a change-of-address database and update your table as required. Another currency concern would be name changes as customers can change their names for various reasons.

Managing data quality



- Determining how to resolve and prevent bad data can be a complex and iterative process. First, you'll implement rules to detect bad data. Then you'll apply those rules to capture and quarantine any bad data. You might need to report any bad data and share the findings with the appropriate domain experts. You and your team can investigate the root cause of each problem, searching for clues upstream in the data lineage. Once you diagnose each problem, you can begin correcting the issues. Ultimately, you want to automate the entire data cleaning workflow as much as possible.

Managing data quality – an example

Scenario: Data from specific data sources consistently has the following data quality issues:

- Missing data (NULL values)

- Out-of-range values

- Duplicate values

- Invalid values

- For example, you need to validate the quality of data in the staging area before loading the data into a data warehouse for analytics. You determine that data from certain data sources consistently has data quality issues including: Missing data, Duplicate values, Out-of-range values, and Invalid values. Here's how an organization might manage and resolve these issues.

Managing data quality - an action plan

Here's how an organization might manage and resolve these issues:

1. Write SQL queries to detect and test for these conditions

Managing data quality - an action plan

Here's how an organization might manage and resolve these issues:

2. Create rules for correcting and managing those conditions

Managing data quality - an action plan

Here's how an organization might manage and resolve these issues:

3. Create a script that runs the data validation SQL queries every night

Managing data quality - an action plan

Here's how an organization might manage and resolve these issues:

4. Automate the script created in step 3 that performs detection and correction processes

Managing data quality - an action plan

Here's how an organization might manage and resolve these issues:

5. Review the automated report and address issues that could not be resolved automatically

- First, write SQL queries to detect these issues and test for them.
- Next, address some of the quality issues that you've repeatedly identified by creating rules for treating them, such as removing rows that have out-of-range values.
- Create a script that runs queries to detect data quality issues that happen during the nightly loads to the data warehouse.
- This script applies corrective measures and transformations for some of these known issues. Next, create a second script that automates the script you

created in step 3.

- After the data is extracted from the various data sources, this script automatically runs the prior script's SQL data validation queries every night in the staging area.
- The script you created in step 3 generates a report of any remaining issues that could not be automatically resolved.
- The administrator can review this report and address the unresolved issues.

Summary

In this video, you learned that:

- Data verification includes checking your data for accuracy, completeness, consistency, and currency
- Data verification is about managing data quality, enhancing data reliability, and maximizing data value
- Determining how to resolve and prevent bad data can be a complex and iterative process
- IBM InfoSphere Information Server for Data Quality can help you perform data verification in a unified environment

What would happen if data quality is ignored?

If data quality is ignored, several negative consequences can arise:

- **Poor Decision-Making:** Decisions based on inaccurate or incomplete data can lead to misguided strategies and actions, potentially harming the organization.
- **Increased Costs:** Organizations may incur additional costs due to errors, such as fixing mistakes, handling customer complaints, or even legal issues arising from incorrect data.

- **Loss of Trust:** Stakeholders, including customers and employees, may lose trust in the organization if they consistently encounter issues related to data quality.
- **Inefficiency:** Time and resources may be wasted on correcting data errors or dealing with the fallout from poor data quality, leading to decreased productivity.
- **Missed Opportunities:** Organizations may miss out on valuable insights or market opportunities because they are working with flawed data.

In summary, ignoring data quality can lead to significant challenges that affect an organization's overall performance and reputation.

How can organizations implement data quality checks effectively?

Organizations can implement data quality checks effectively by following these steps:

1. Define Data Quality Standards:

- Establish clear criteria for what constitutes high-quality data, including accuracy, completeness, consistency, and currency.

2. Automate Data Validation:

- Use automated tools and scripts to regularly check data against the defined standards. This can help identify issues quickly.

3. Implement Data Profiling:

- Analyze existing data to understand its structure, content, and quality. This helps in identifying patterns and potential problems.

4. Create Data Quality Rules:

- Develop specific rules for handling common data quality issues, such as removing duplicates or flagging out-of-range values.

5. Establish a Data Governance Framework:

- Assign roles and responsibilities for data quality management within the organization. This ensures accountability and oversight.

6. Train Employees:

- Provide training for staff on the importance of data quality and how to maintain it. This fosters a culture of data responsibility.

7. Monitor and Report:

- Continuously monitor data quality and generate reports on findings. This helps in tracking improvements and identifying ongoing issues.

8. Iterate and Improve:

- Regularly review and update data quality processes based on feedback and changing business needs.

By implementing these steps, organizations can enhance their data quality and ensure that their data is reliable for decision-making.

How can you automate data validation processes?

Automating data validation processes can be achieved through several methods:

1. Use of Data Quality Tools:

- Implement specialized data quality software (e.g., IBM InfoSphere, Talend) that offers built-in validation features to automatically check data against predefined rules.

2. SQL Queries:

- Write SQL scripts to run validation checks on databases. For example, you can create queries to identify duplicates, check for null values, or validate data formats.

3. Data Pipelines:

- Set up data pipelines using ETL (Extract, Transform, Load) tools that include validation steps. These pipelines can automatically validate data as it moves from source to destination.

4. Scheduled Jobs:

- Use scheduling tools (like cron jobs) to run validation scripts at regular intervals, ensuring that data is continuously monitored for quality.

5. Data Profiling:

- Implement data profiling techniques that automatically analyze data sets to identify anomalies, patterns, and quality issues.

6. Alerts and Notifications:

- Set up automated alerts to notify relevant personnel when data quality issues are detected, allowing for quick resolution.

7. Integration with APIs:

- Use APIs to integrate data validation checks into existing applications, ensuring that data is validated in real-time as it is entered or processed.

By leveraging these methods, organizations can streamline their data validation processes, reduce manual effort, and improve overall data quality.